Veritas: Claim Verification & Bias Detection

Lightweight Chrome Extension to Analyze Text for Political Bias and Veracity

Scan to Github

Team Members

Madhav Bezawada, Divij Dhiraaj, Louis Doan, Mustakim Kazi, Keyu Lin, Haneef Taher

Supervisor

Dr. Allison Lahnala | McMaster University, CAS

Objective

Misinformation and bias are present in many online spaces, such as news articles, blogs, and forums. The purpose of Veritas is to provide everyday people the ability to quickly and easily analyze online passages of text for factuality and bias.

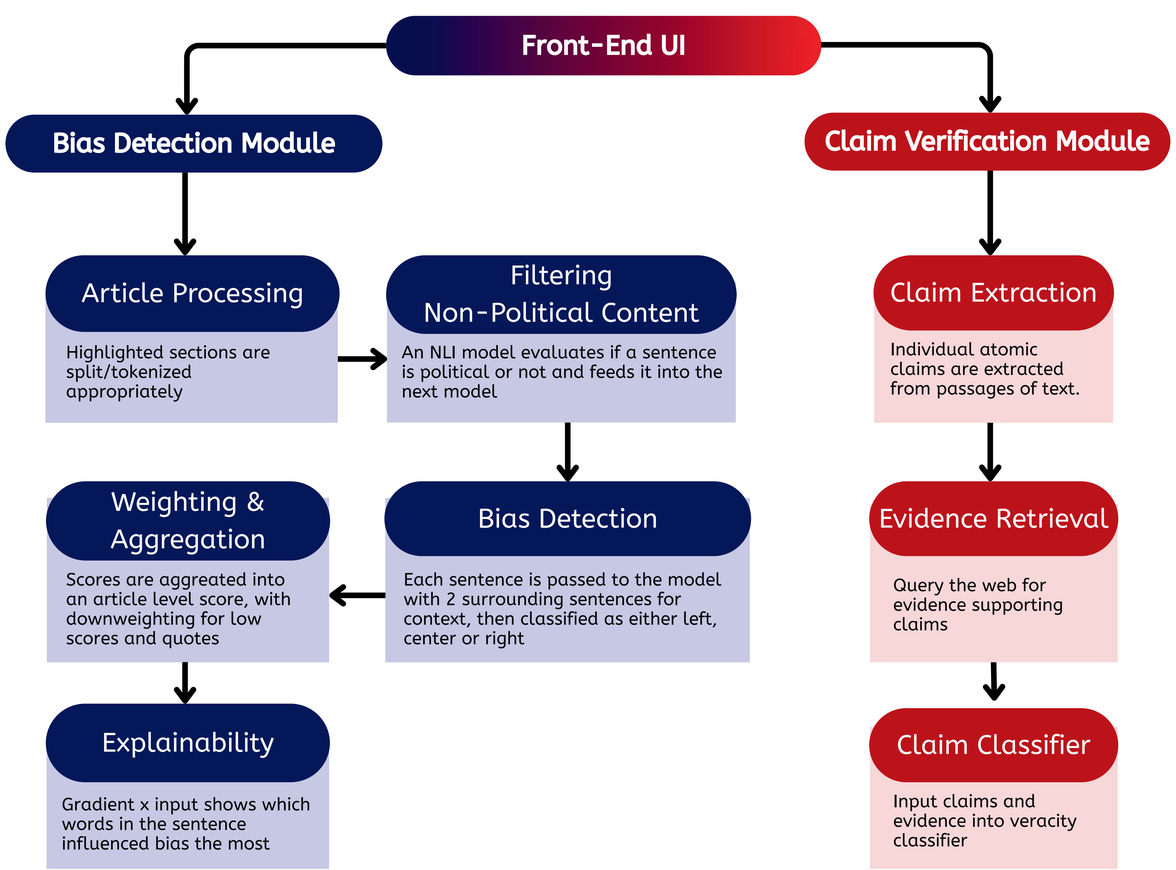
Tech Stack

FastAPI

mongoDB

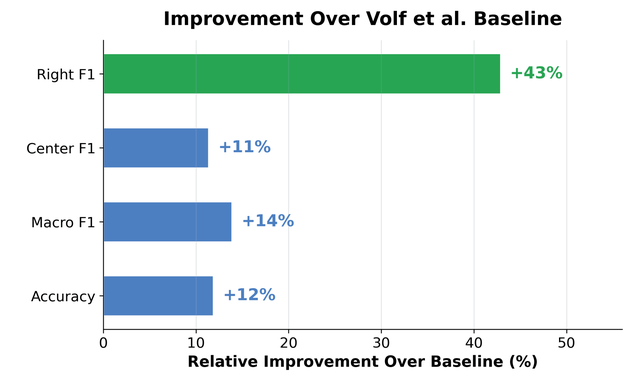
TS

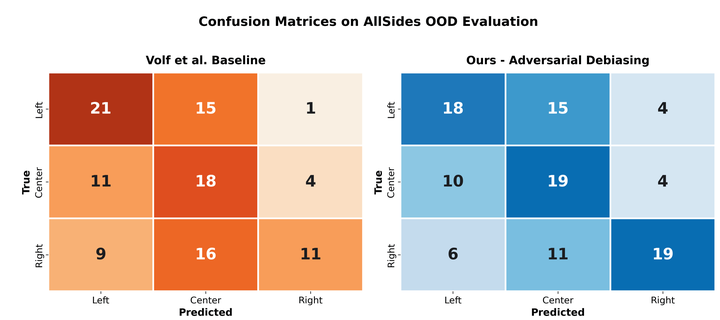
Methodology



Bias Detection Benchmarks

We evaluated both models on 106 AllSides articles from 2026, completely unseen during training. This is a harder test than standard cross-dataset evaluation, the Volf et al. baseline drops from 66.2% F1 on their leave-one-out to 47.2% on our benchmark. Using adversarial debiasing with gradient reversal (Baly et al., 2020) on DeBERTa-v3-large, we improved accuracy by 12%, macro F1 by 14%, and right-leaning detection by 43%.





More Bias Benchmarks

The baseline model shows a strong left-leaning bias, correctly classifying only 11 of 36 right-leaning articles while misclassifying 16 as center. Our adversarial debiasing trains a gradient reversal layer that forces the encoder to learn an ideological signal rather than source identity, producing a more balanced confusion matrix.

Claim Classification Benchmarks

Comparing our baseline model to our FEVER trained fine-tuned model, we evaluated on the AveriTeC dataset to determine the effectiveness of each model. We can see that our model improves on the majority of metrics including the most important aggregated metrics such as F1, macro-average and weighted-average. While our model does not perform well with "Not Enough Info" labeled claims, this is a problem with NLP classifiers evidenced by the baseline's similar poor performance where it is hard to determine insufficient evidence for a claim classification.

Claim Verification Metrics

	Precision(Baseline)	Precision(Model)	Recall(Baseline)	Recall(Model)	F1(Baseline)	F1(Model)	Support
Refuted	0.83	0.77	0.51	0.92	0.63	0.84	117
Not Enough Info	0.14	0.08	0.58	0.08	0.23	0.14	24
Supported	0.55	0.62	0.15	0.56	0.23	0.59	41
Accuracy					0.44	0.73	182
Macro Average	0.51	0.63	0.41	0.52	0.36	0.52	182
Weighted Average	0.68	0.7	0.44	0.73	0.49	0.69	182